

Nuclear Energy (Phoebe)

1. What Are New Jersey's Current Plans for Nuclear Energy?

- Nuclear energy is the single largest source of carbon-free electricity in NJ, supplying approximately 39%–41% of total statewide electricity generation.
- The state currently relies on two nuclear power stations in Salem County:

Salem Nuclear Generating Station (Units 1 & 2)

Hope Creek Nuclear Generating Station (Unit 1)

All three reactors are owned and operated by PSEG Nuclear LLC in partnership with Exelon Generation(for Salem).

- NJ's 2019 Energy Master Plan and updated 2024 Clean Energy Strategy affirm nuclear as critical to meeting the state's goal of 100% clean electricity by 2035.
- No new reactors are planned or under construction, but NJ's policy is to preserve and extend the life of its existing nuclear assets.
- To achieve this, NJ has implemented the Zero Emission Certificate (ZEC) program, which provides ratepayer-funded subsidies to keep nuclear plants economically viable despite competition from cheaper fossil fuels and renewables.

The program was renewed in 2022 for an additional three-year cycle and will be reviewed again before 2025.

2. Development Plans & Energy Industry Participation

- No plans for new nuclear construction currently exist due to:

High upfront capital costs

Lengthy permitting timelines

Public concern around radioactive waste and safety

- However, there is state and industry interest in emerging technologies like Small Modular Reactors (SMRs) and advanced reactor designs that may be viable in the future.
- The NJ Board of Public Utilities (NJBPU) has supported research and stakeholder outreach related to advanced nuclear in grid reliability discussions.
- PSEG Nuclear, the primary nuclear operator in NJ, continues to:

Invest in safety and performance upgrades

Participate in DOE-led resilience and innovation initiatives

Advocate for continuation of the ZEC program

- The regional power grid operator, PJM Interconnection, has emphasized the need for firm clean baseload like nuclear to balance intermittent wind and solar resources.

3. Federal Incentives – "Big Beautiful Bill" (IRA) and Prior Support

- Under the Inflation Reduction Act (IRA, 2022):

Existing nuclear plants qualify for a new Production Tax Credit (PTC) of up to \$15/MWh to ensure they remain profitable in low-cost power markets.

Bonus tax credits are available for plants that retain union workers or source components domestically.

While NJ is not building new nuclear facilities, existing plants like Salem and Hope Creek can benefit directly from this financial support.

- The Bipartisan Infrastructure Law (BIL, 2021):

Funds R&D for advanced nuclear and grants for grid integration

DOE launched the Civil Nuclear Credit Program (\$6 billion) to prevent early closure of nuclear plants across the country

- Past federal support in NJ:

NJ's ZEC program was designed in part based on federal guidance about compensating carbon-free power.

NJ has previously partnered with the DOE on nuclear plant cybersecurity, safety enhancements, and energy modeling studies.

4. Recent NJ Legislation and Budget Actions (as of July 2025)

- No new laws passed in July 2025 specifically addressed nuclear energy, but related legislation supports nuclear indirectly:

Bill S-4289, signed into law, mandates procurement of 1,000 MW of grid-scale energy storage by 2026, helping nuclear plants operate more efficiently by balancing supply during off-peak periods.

A new law directs NJBPU to study the impact of data centers on grid demand, a growing sector that may increase reliance on firm baseload like nuclear.

- Budget concern: Lawmakers authorized a \$190 million transfer from the Clean Energy Fund to NJ Transit, the second-largest diversion in the fund's history.

While this doesn't directly cut nuclear funds, it weakens overall clean energy financing and may impact the balance of renewable-nuclear grid integration efforts.